nature chemical biology

Article

https://doi.org/10.1038/s41589-022-01223-z

Co-evolution-based prediction of metal-binding sites in proteomes by machine learning

Received: 11 January 2022

Accepted: 8 November 2022

Published online: 2 January 2023

Check for updates

Yao Cheng[1,2,7], Haobo Wang[1,2,7], Hua Xu[3,7], Yuan Liu[1,2,7] ✉, Bin Ma[1,2], Xuemin Chen[1,2], Xin Zeng[4], Xianghe Wang[1,2], Bo Wang[3], Carina Shiau[5], Sergey Ovchinnikov[6], Xiao-Dong Su[3] ✉ & Chu Wang[1,2,4] ✉

Metal ions have various important biological roles in proteins, including structural maintenance, molecular recognition and catalysis. Previous methods of predicting metal-binding sites in proteomes were based on either sequence or structural motifs. Here we developed a co-evolution-based pipeline named 'MetalNet' to systematically predict metal-binding sites in proteomes. We applied MetalNet to proteomes of four representative prokaryotic species and predicted 4,849 potential metalloproteins, which substantially expands the currently annotated metalloproteomes. We biochemically and structurally validated previously unannotated metal-binding sites in several proteins, including apo-citrate lyase phosphoribosyl-dephospho-CoA transferase citX, an *Escherichia coli* enzyme lacking structural or sequence homology to any known metalloprotein (Protein Data Bank (PDB) codes: 7DCM and 7DCN). MetalNet also successfully recapitulated all known zinc-binding sites from the human spliceosome complex. The pipeline of MetalNet provides a unique and enabling tool for interrogating the hidden metalloproteome and studying metal biology.

Metal ions have indispensable roles in many biological processes. Owing to their unique physical and chemical properties, metal ions are frequently exploited to stabilize protein structures and as cofactors in proteins to aid catalysis of biochemical reactions and transductions[1]. It has been estimated that more than one-third of the entire proteomes are metal-binding proteins; however, given that only a few amino acid residues are normally involved in metal binding, it is not an easy task to generally predict metal-binding proteins[2]. Discovery and functional characterization of new metal-binding proteins in proteomes will therefore be of great interest for both basic biology and industrial applications in the postgenome era.

To hold specific metal ions in a stable and energetically favorable manner, metalloproteins across all species have evolved topologically similar metal-binding sites in their structures to meet the chemical and geometric bonding requirements[3]. Over the decades, both structural observations and theoretical calculations have accumulated detailed knowledge to accurately describe the rules of binding geometry and valence for different metal ions[4]. More recently, researchers have

[1]Synthetic and Functional Biomolecules Center, Beijing National Laboratory for Molecular Sciences, Key Laboratory of Bioorganic Chemistry and Molecular Engineering of Ministry of Education, Peking University, Beijing, China. [2]Department of Chemical Biology, College of Chemistry and Molecular Engineering, Peking University, Beijing, China. [3]State Key Laboratory of Protein and Plant Gene Research, and Biomedical Pioneering Innovation Center (BIOPIC), Peking University, Beijing, China. [4]Peking-Tsinghua Center for Life Sciences, Peking University, Beijing, China. [5]Cornell University, Ithaca, NY, USA. [6]John Harvard Distinguished Science Fellow, Harvard University, Cambridge, MA, USA. [7]These authors contributed equally: Yao Cheng, Haobo Wang, Hua Xu, Yuan Liu. ✉e-mail: wendao@pku.edu.cn; xdsu@pku.edu.cn; chuwang@pku.edu.cn

explored metal-binding proteins directly in proteomes using other experimental methods including inductively coupled plasma mass spectrometry (ICP-MS)[5], activity-based protein profiling[6] and metal isotope native radio autography in gel electrophoresis[7].

In light of the explosive growth of genome databases, various bioinformatic methods have also been developed to predict metal-binding sites based on the sequence or/and structural information. For example, zinc-binding proteins in human proteome have been systematically predicted by a combination of sequence and structural homology to known metalloproteins[8]. More sophisticated algorithms such as support vector machines and neural networks were used to identify cysteines and histidines in transition-metal-binding sites from protein primary sequences[9–11]. Moreover, knowledge of canonical chelating geometries could also be utilized to predict metal-binding sites from apoprotein structures[12]. Lastly, web servers such as MIB[13] and COFACTOR[14] have been developed to enable automated annotation for metal-binding sites or ligand-binding sites in proteins[14]. However, it still remains challenging to computationally discover new classes of metalloproteins without sequence or structural homology in proteomes[3].

Thanks to high-throughput sequencing technologies, we are now having access to genome information for a rapidly growing list of species. Alignment of sequences for a given protein of interest (POI) from multiple species enables the construction of its comprehensive evolution history, containing not only highly conserved residues but also pairs of residues with strong covariances. Such co-evolution information, which is often represented in the so-called 'contact map', has been extracted for precise detection of long-range interactions in sequence[15–20]. More recently, co-evolution signals were harnessed to enable landmark successes in structural predictions for globular proteins[21–23], protein complexes[24,25] and structured states of disordered regions[26]. Co-evolution analyses have also been implemented to define functional sites in proteins in several approaches[27–30]; however, its use for systematic prediction of metal-binding sites has not been explored.

Here we have developed a machine-learning (ML) method named 'MetalNet' to predict metal-binding proteins in proteomes with site-specific resolution using co-evolution. We applied MetalNet to several representative prokaryotic proteomes and predicted nearly 5,000 potential metalloproteins. We biochemically and structurally validated several MetalNet predictions that were previously unannotated, including one zinc-binding site from the apo-citrate lyase phosphoribosyl-dephospho-CoA transferase citX. In addition, MetalNet also successfully predicted all known zinc-binding sites from the human spliceosome complex. This unique computational pipeline will provide a useful tool for exploring unknown metalloproteomes and studying metal biology.

Results

Prediction of co-evolved metal-binding residue pairs

We first collected metalloproteins from the Protein Data Bank (PDB) and crosschecked with the sequence culling database PISCES[31] to get a list of metalloproteins[32]. We also extracted the contact maps for these proteins using the MSA-transformer model (Methods). Classification of the metal-chelating residues showed that cysteine (C), histidine (H), glutamic acid (E) and aspartic acid (D) (together referred as 'CHED') appear much more frequently in metal-binding sites, accounting for 91% of all the metal-chelating residues (Fig. 1a). For simplicity, we focused only on these four types of amino acids as potential metal-binding residues for following analyses.

We next analyzed co-evolution signals from all the ten possible pairwise combinations from CHED and observed that many of them showed enriched co-evolution signals compared to those pairs randomly picked from proteomes (Fig. 1b). For example, 53% of the co-evolved CC residue pairs are involved in metal binding (Fig. 1b), which is tenfold more enriched than a random CC pair if considering that only 23% of the cysteines in proteomes are chelating metals (Fig. 1c). The results suggested that we could in principle incorporate the co-evolution information into a prediction model to differentiate metal-binding residue pairs from nonbinding ones.

To systematically formulate this approach, we generated a 'pairwise frequency matrix' for every CHED co-evolved residue pair in the contact map. The 21 × 21 matrix is calculated based on the residue–residue frequency profile observed in the homologous sequences for this given pair (20 amino acids and one more position gap). We averaged the matrices for all the metal-binding and nonbinding residue pairs, respectively, and observed very distinct patterns between them (Fig. 1d). For example, the metal-binding pairs have much more enriched signals in the CHED region whereas signals for the nonbinding pairs are dispersed throughout the whole matrix. We reasoned that such pattern differences can be transformed into a classification problem that is addressable by ML.

We compiled a nonredundant list of 697 metalloproteins from the aforementioned dataset that cover ten types of metals (ZN, CA, MG, MN, FE, SF4, NI, CU, CO and FES in Extended Data Fig. 1) and defined a total of 2,346 co-evolved metal-binding CHED pairs in these proteins as the positive samples (Supplementary Dataset 1). We also defined another set of 16,816 co-evolved but nonmetal-binding CHED pairs from the same set as the negative samples (Supplementary Dataset 1). We calculated the pairwise frequency matrices of every pair and used AutoGluon (https://auto.gluon.ai/stable/index.html) to train several ML models and evaluated the performance with the 5-fold cross-validation test. The final model exhibited the performance with an area under the Receiver Operating Characteristic Curve (AUC) of 0.88, F1 score of 0.66 and Matthews Correlation Coefficient (MCC) of 0.63 (Supplementary Table 1).

Assembling co-evolved metal-binding pairs into network

During investigating the co-evolution network of proteins in the benchmark, we observed that many metal-binding co-evolved pairs assemble to form an integrated network (Extended Data Fig. 2). This is consistent with a previous report that residues within a functional site have the tendency to cluster together in a co-evolution network[29]. To quantify whether metal-binding residues are more frequently engaged in a connected co-evolution network, we calculated the number of co-evolved connections ('node degree') for every CHED residue. The results showed that the metal-binding residues indeed have higher node degrees on average than the nonbinding ones (Extended Data Fig. 3). We reasoned that detecting such networks should help improve the precision of our prediction (Fig. 2). We therefore implemented a graph-based approach to detect co-evolution network clusters from the predicted metal-binding residue pairs by trimming off those 'singleton' pairs (Methods). As expected, applying the graph filter resulted in an improved F1 score of the prediction (Supplementary Table 2).

To recover certain sites with fewer sidechain liganding residues such as Mg/Ca-binding sites (Extended Data Fig. 4), we added a 'repechage' rule that when there are no residues left after applying the graph filter, we would keep the top *N* pair with the highest predicted possibility from those which do not pass the filter. We systematically evaluated the F1 scores with *N* varying from 1 to 5 and the results showed that recovery of the single top-ranking pair by the repechage rule yielded the best F1 score (Supplementary Table 2). We therefore settled on this parameter and assigned the best-predicted pair as potential metal-binding sites if no co-evolved networks are detected.

Interestingly, among the extracted co-evolution networks, we also observed different network patterns for sites with different types of metal bound (Fig. 2). We compiled such patterns from known metalloproteins to construct a metal-type-dependent architecture database of co-evolution networks ('co-evolution motif bank'; Supplementary Dataset 2). Because different metal ions have their preferred co-evolution network architecture, it should be possible to compare

https://doi.org/10.1038/s41589-022-01223-z

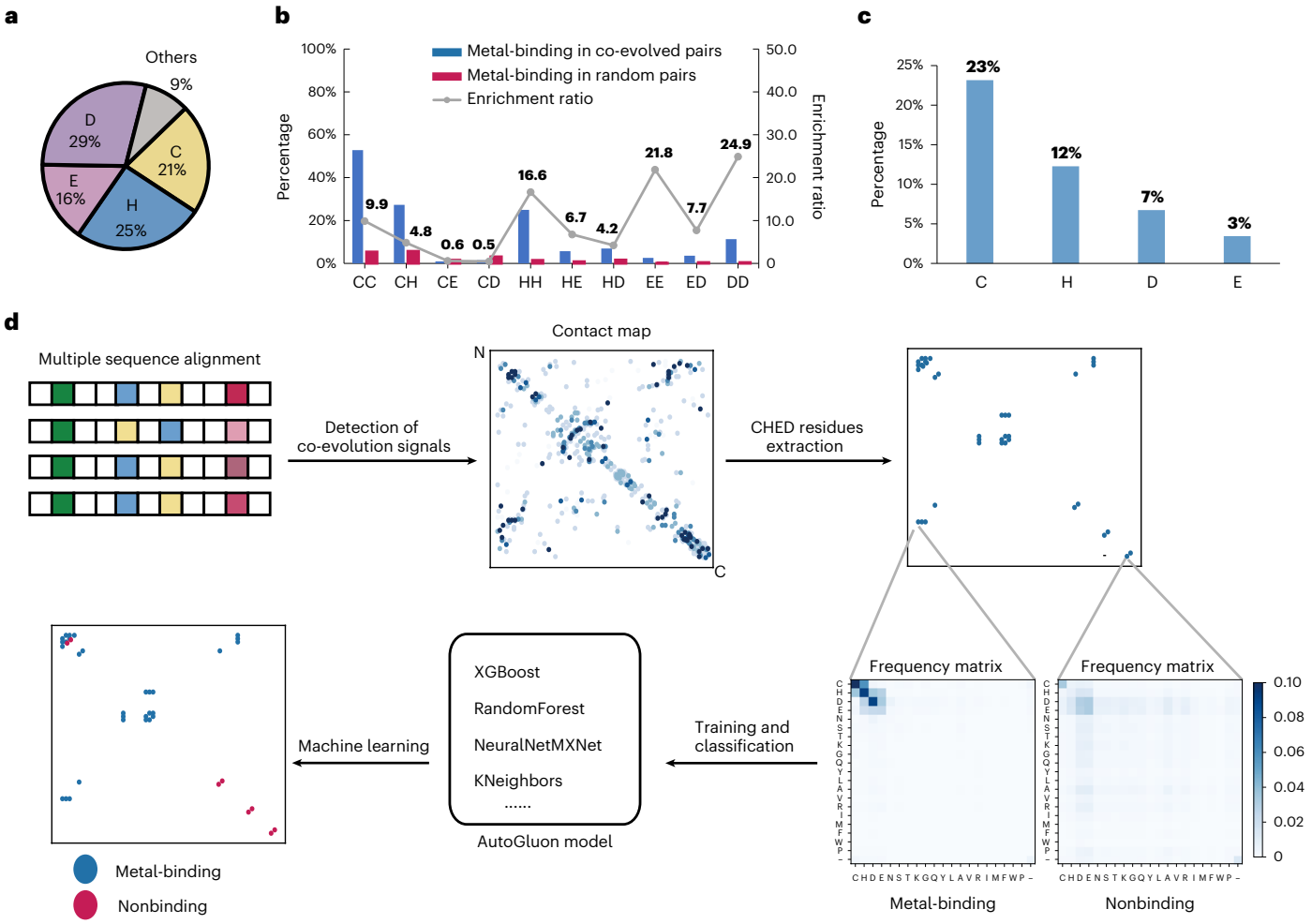


Fig. 1 | Predicting metal-binding residue pairs from co-evolution by ML. **a**, Classification of metal-chelating residues by amino acid types, showing CHED accounting for 91% of all the metal-chelating residues. **b**, Co-evolved residue pairs are more enriched with metal binding. For each of the ten CHED pairwise combinations, the percentage of co-evolved pairs observed with metal binding (blue bar) and the percentage of metal-binding pairs in randomly picked residue pairs (red bar) were calculated. Gray line shows the enrichment ratios. **c**, Percentage of CHED residue type, which is involved in metal-binding as observed in the training set. **d**, Scheme of predicting metal-binding residue pairs based on the co-evolution information using ML. Starting with MSA, all CHED co-evolved pairs were extracted from the contact map and their pairwise residue-composition frequency matrices were used to train a model to predict whether a co-evolved pair is metal-binding for every CHED co-evolved residue pair in the contact map. The 21 × 21 pairwise frequency matrix is calculated based on the residue–residue frequency profile observed in the homologous sequences.

the residue compositions and topology of a predicted co-evolution network with existing patterns in the co-evolution motif bank and assign the possible type of metal ion bound (Fig. 2).

Overall workflow of MetalNet

We combined the ML-based classifier and the graph-based network detection to streamline the workflow of our method, named 'MetalNet' (Extended Data Fig. 5). For a POI, we first obtained its multiple sequence alignment (MSA) and calculated its co-evolution contact. Next, we extracted all co-evolved CHED residue pairs from the contact map, and for each pair, we fed its pairwise frequency matrix to the ML model to determine whether it binds metals. We proceeded to apply the graph filter to detect connected sub-network clusters and keep one pair with the highest predicted possibility by ML model if no pairs are left after graph filter (Extended Data Fig. 5). As a result, we obtain one or more co-evolution clusters within the POI, each of which represents a potential metal-binding site. Whenever possible, we use the residue composition and topology information of the network cluster to aid annotation of the type of metal ion bound in this specific site. Overall, the method needs only the MSA-derived co-evolution information as the input and does not rely on any sequence or structural motifs to make predictions.

We briefly compared the performance of MetalNet with a well-known structure-based prediction method MIB[13]. We collected metalloproteins that have been released recently in PDB so that they are 'new' to both MetalNet and MIB and used them as an unbiased testing dataset (Supplementary Dataset 3). Overall, MetalNet shows better precision and MIB has a better recall, resulting in comparable F1 score performance between them (Extended Data Fig. 6). Considering that these methods take sequence co-evolution and structure information as input, respectively, it will be worth integrating their features together to enhance the prediction power in future.

Prediction of metal-binding sites in prokaryotic proteomes

We applied MetalNet to predict metalloproteins in four prokaryotic proteomes, including *Escherichia coli*, *Bacillus subtilis*, *Saccharolobus solfataricus* and *Halobacterium salinarum*, which currently have 1,965 annotated metalloproteins. MetalNet predicted a total of 4,849 potential metalloproteins in these species (Fig. 3a and Supplementary Dataset 4). Moreover, MetalNet found 1,603 network clusters and

https://doi.org/10.1038/s41589-022-01223-z

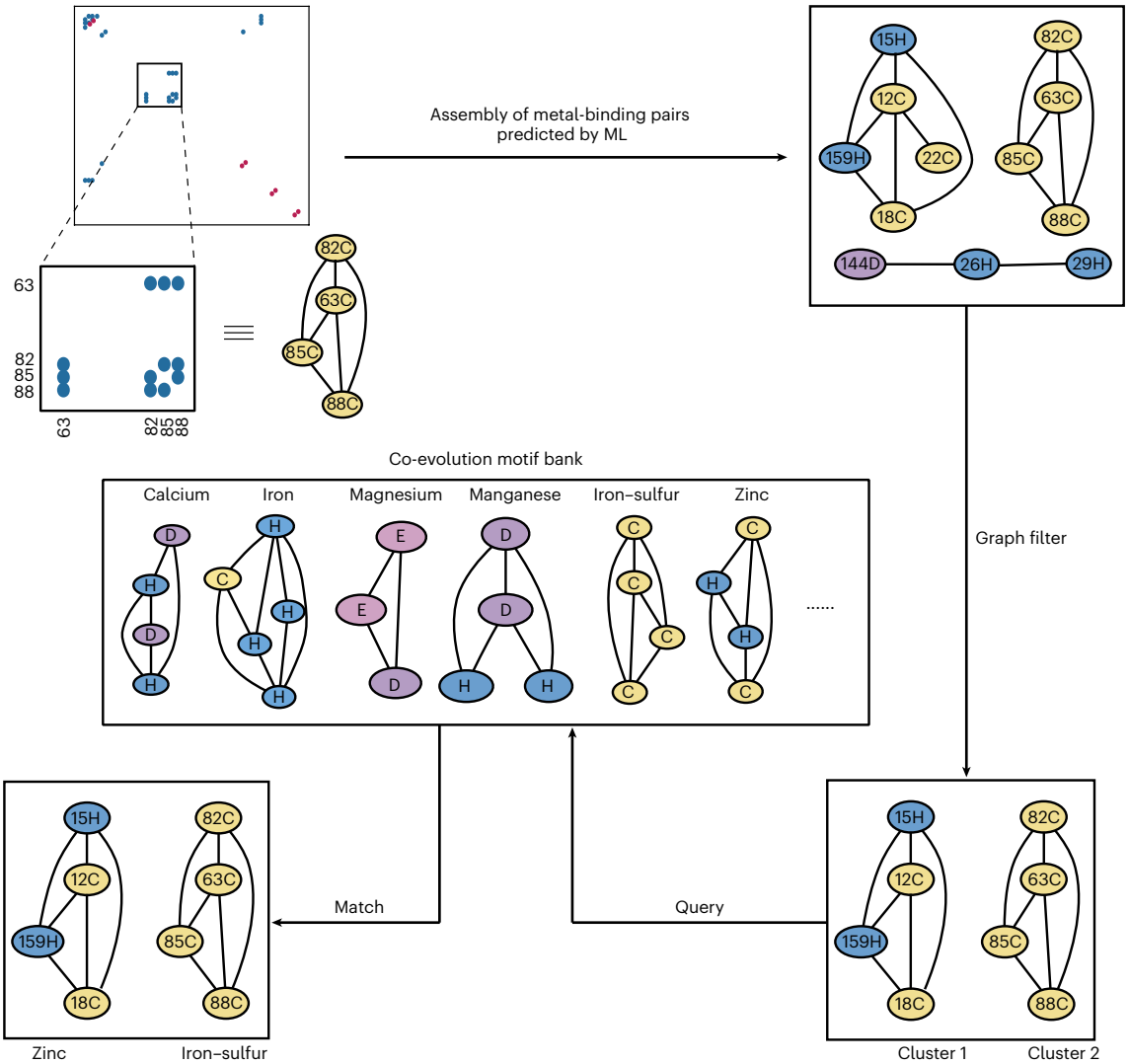


Fig. 2 | Assembly of metal-binding residue pairs into a high-order connected co-evolution network by graph theory. Co-evolved metal-binding residue pairs predicted by ML model are assembled into a high-order co-evolution network, and the connected sub-networks are then detected by graph theory. The 'singleton' pairs are discarded, which can serve as a filter to increase the prediction precision. Finally, the predicted sub-network clusters are queried in a 'co-evolution motif bank' compiled from known metalloproteins (histidine in blue, cysteine in yellow, aspartic acid in purple and glutamic acid in pink), which will enable prediction of the type of metal ions bound. It should be noted that one metal type might have multiple corresponding network motifs in the bank.

we were able to match the co-evolution network patterns for 973 of them so that their metal types could be inferred by analogy to existing metalloproteins (Supplementary Dataset 4).

We first sought direct structural evidence to support our predictions. For each of the predicted metalloproteins, we searched its homologs by PDB-BLAST[33] and found that 37% (1,810 of 4,849) of the predictions could be supported by experimental structures (Fig. 3b). Among these predictions, we not only observed classical short sequence motifs such as C2H2 in zinc fingers and CXXCXXCC in certain iron–sulfur-binding proteins but also found cases with metal-chelating residues by >100 amino acids apart in sequence (Fig. 3c) or with complexed co-evolved networks corresponding to multinuclear or multiple metal-binding sites (Fig. 3d,e). Collectively, these examples highlighted the power of MetalNet in discovering complex metal-binding sites from proteins independent from 1D sequences or 3D structural motifs.

For the rest of the predicted proteins that do not have direct structural evidence, we searched for indirect but relevant evidence from various databases to further support our predictions[1,8], including UniProt, Gene Ontology and Pfam (Fig. 3b). Collectively, an additional 445 predicted proteins can be supported by at least one piece of this indirect evidence (Fig. 3b), accounting for 9% of the total predictions. Lastly, there are 2,594 predicted proteins that have never been annotated previously as metalloproteins (Supplementary Dataset 4). We envision that these MetalNet predictions will greatly facilitate functional annotations of uncharacterized metalloproteomes in these species.

Prediction from a metagenome-enhanced Pfam database

We next investigated whether evolutionary information extracted from metagenome sequencing data can enhance MetalNet predictions. We started with the MSAs that were previously calculated by GREMLIN to guide the structural modeling of 1,116 Pfam families by Rosetta (ref. [22]). MetalNet predicted 175 metalloproteins with clusters and 60 of them have overlapping Rosetta models (Fig. 4a and Supplementary Dataset 5). Analysis of these models showed that residues within 46 predicted metal-binding sites are proximal to each other (Supplementary Dataset 5 and Supplementary Fig. 1). Furthermore, the overall distribution of

https://doi.org/10.1038/s41589-022-01223-z

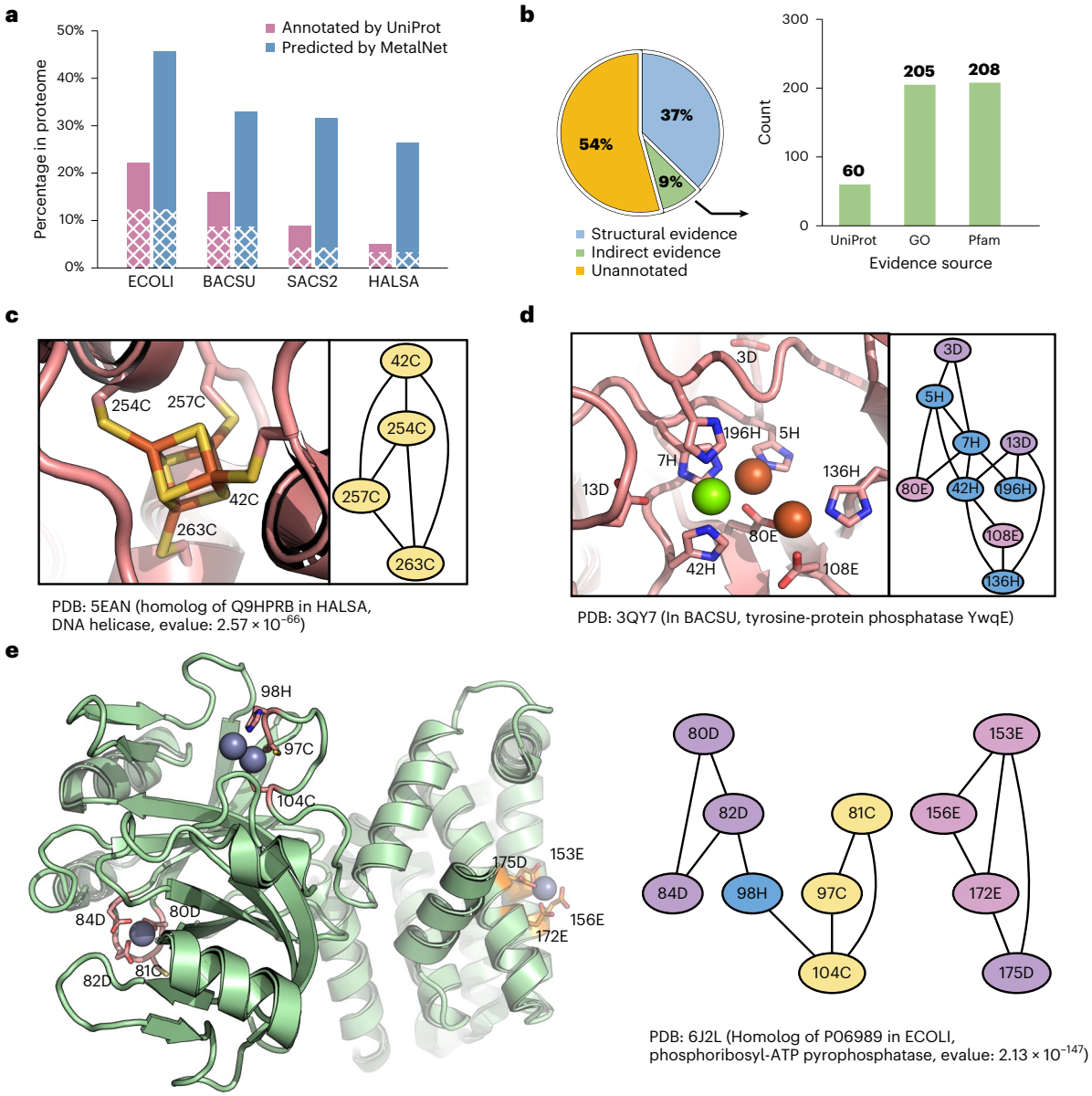


Fig. 3 | Predictions and analyses of metalloproteins from representative prokaryotic species. a, The percentage of predicted metalloproteins by MetalNet (blue) and annotated proteins by UniProt (pink) within each proteome. The overlapping proteins are highlighted in the white-patch pattern. ECOLI: *E. coli*; BACUS: *B. subtilis*; SACS2: *S. solfataricus*; HALSA: *H. salinarum*. **b**, Part of MetalNet predictions can be supported by either direct structural evidence or indirect evidence from various sources, which include UniProt annotations, Gene Ontology annotations and Pfam annotations. **c**, Example of a predicted mononuclear iron–sulfur cluster binding protein from *H. salinarum*, which is supported by its structural homolog (PDB code: 5EAN). **d**, Example of a predicted multinuclear site in YwqE from *B. subtilis*, which is supported by the structure in PDB (code: 3QY7). **e**, Example of three separate zinc-binding sites, which is supported by its structural homolog (PDB code: 6J2L). In **c**–**e**, crystal structures of corresponding metalloprotein are shown in cartoon and the predicted metal-binding sites are shown as network. Metal-chelating residues are shown in sticks and metals bound are shown in either sticks (iron–sulfur cluster in yellow and brown) or spheres (magnesium in green, iron in brown and zinc in gray).

residue–residue distances was consistent with that calculated from known metalloproteins below 4.5 Å (Extended Data Fig. 7).

We selected 21 MetalNet predictions for experimental validation, none of which has been reported with any metal-binding activities or homologs back then at that time. Five targets yielded reasonable expression in the initial screening including Rv3033, Desor_0198, SVEN_5263, cbrC and citX. Among them, Rv3033 (Uniprot code: I6YAY5) failed to be purified likely because it is predicted as an iron–sulfur cluster-binding protein. Purified Desor_0198 (UniProt code: G7W7J8) and SVEN_5263 (UniProt code: F2R5E5) showed zinc-binding signals by ICP-MS (Extended Data Fig. 8a,b); however, their metal-binding site variants were unstable for purification. We therefore characterized the remaining two proteins, cbrC and citX, in detail by biochemical and/or structural methods.

The protein cbrC (UniProt code: P31469, Pfam: PF03691) is a CreB-regulated gene C protein in *E.coli*[34]. MetalNet predicted two potential metal-binding sites (Site 1: C31, C33, C34, C56 and C59; Site 2: H125, C126, C182 and C185) (Extended Data Fig. 8c). We purified the wild-type protein, the single variant of C56S or C182S, as well as the double variant (Extended Data Fig. 8c). ICP-MS analysis showed that the wild-type protein was bound with zinc and the measured metal to protein ratio was close to two, which is highly consistent with the MetalNet prediction. The zinc/protein ratios of the single variants, C56S and C182S, were measured as 0.481 and 0.628, respectively, and

https://doi.org/10.1038/s41589-022-01223-z

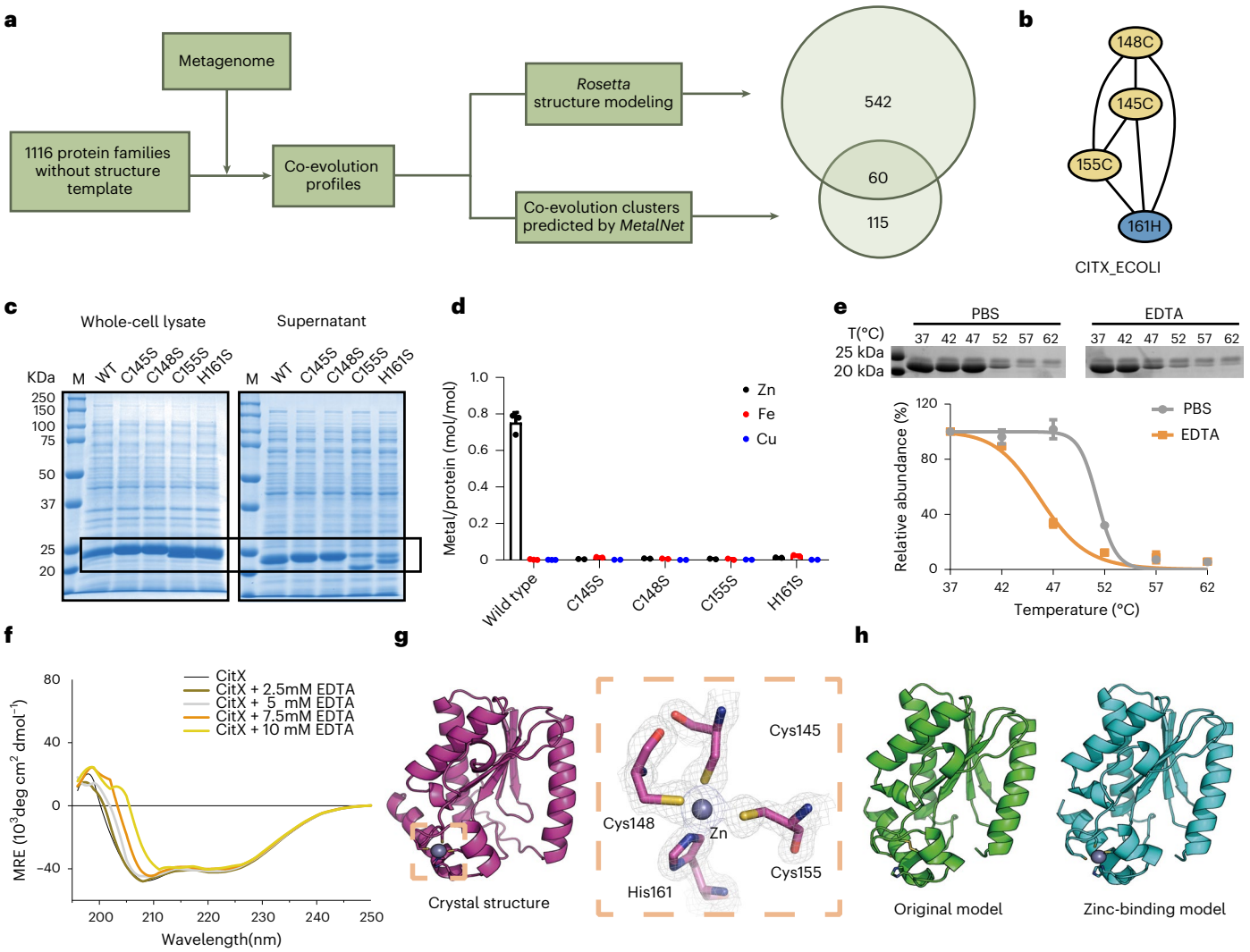


Fig. 4 | Biochemical and structural characterization of citX as a zinc-binding protein. a, Scheme of predicting metalloproteins from the Pfam families that lack existing structural homologs using metagenome-enhanced co-evolution. **b**, The predicted zinc-binding co-evolved network in citX by MetalNet. **c**, Variant of some of the predicted metal-binding residues compromised citX's solubility. The wild-type and all four variants (C145S, C148S, C155S and H161S) showed similar protein abundance as demonstrated by Coomassie blue staining (highlighted in the black box). The experiment was repeated twice independently with similar results. **d**, Variant of the predicted metal-binding residues abolished the zinc-binding in citX as measured by ICP-MS. **e**, Treatment of EDTA, a zinc ion chelator, decreased citX's thermal stability as measured by the thermal shift assay. In **d** and **e**, error bars represent mean and s.d. ($n = 3$ biologically independent samples) **f**, Circular dichroism measurement of the purified citX in presence of various concentrations of EDTA. **g**, Crystal structure of citX (PDB code: 7DCN) and the zoom-in view of the zinc-binding site confirms the prediction of MetalNet. **h**, Structural models of citX built by Rosetta with (right) and without (left) the predicted zinc-binding site into consideration.

for the double variant of C56S/C182S, its metal-binding activity was completely abolished (Extended Data Fig. 8c).

The protein citX (UniProt code: P0A6G5, Pfam: PF03802) was annotated as an apo-citrate lyase phosphoribosyl-dephospho-CoA transferase in *E. coli*[35]. MetalNet identified a co-evolution network consisting of three cysteines (C145, C148 and C155) and one histidine H161 in citX (Fig. 4b), and the network topology confidently matches a zinc-binding site. We recombinantly overexpressed the wild-type citX and the single variants of the four metal-binding residues (C145S, C148S, C155S and H161S) in *E.coli*, and some of the single variants showed compromised solubility (Fig. 4c). After purification (Extended Data Fig. 8d), ICP-MS analysis confirmed that the wild-type protein specifically bound zinc but not iron or copper as expected from the prediction, and the measured metal to protein ratio was around 0.7 and none of the single variants showed detectable zinc-binding signals (Fig. 4d). When the wild-type protein was treated with EDTA, a conventional metal-chelator, the thermal stability curve was shifted left compared with the sample treated with PBS, indicating the protein was destabilized upon metal extraction (Fig. 4e). Circular dichroism analysis also showed that EDTA treatment would greatly perturb the protein's conformation (Fig. 4f). Finally, we solved the structure of citX by X-ray crystallography (PDB code: 7DCM at 2.5 Å resolution determined by single-wavelength anomalous diffraction and PDB code: 7DCN at 1.7 Å resolution determined by molecular replacement; Supplementary Table 3). In the structure of 7DCN, distances between the zinc ion and the predicted chelating residues are 2.34 Å, 2.31 Å, 2.36 Å and 2.08 Å, which unambiguously proved that the predicted cysteines and histidine together form a tetrahedral zinc-binding site in citX (Fig. 4g). The crystal structure also matches well with a structural model we built by Rosetta (Fig. 4h). To investigate structural and dynamic characteristics of citX and its variants, we also performed molecular dynamic (MD) simulations and the results showed that all the four

variants had greater conformational fluctuations at the zinc-binding site, suggesting that the metal ion has a stabilizing effect for the protein (Extended Data Fig. 9).

Prediction of metalloproteins from human spliceosome

One major challenge of applying MetalNet to eukaryotic proteomes is the relatively poor quality of MSA due to the low depth of sequence information that is currently available from a limited number of eukaryotic species. To demonstrate that MetalNet is generally applicable to any type of proteome as long as the target protein has MSA with reasonable quality, we, as a proof of principle, attempted to apply MetalNet to the human spliceosome because the giant protein machinery is functionally conserved across almost all eukaryotic organisms[36] and multiple structures have been solved[37,38] (Extended Data Fig. 10). We utilized deepMSA[39] to generate high-quality MSA from available sequences and applied MetalNet to finally predict 34 potential metal-binding sites from 33 spliceosome subunits (Supplementary Dataset 6). Notably, all zinc-binding sites that were observed in the experimental structures could be recapitulated by MetalNet (Extended Data Fig. 10) and other unannotated metal-binding sites remained biochemically and functionally validated.

Discussion

Generally speaking, co-evolved residues are likely to be closed in the 3D structure and therefore the co-evolution information has been extensively used in protein structure prediction. MetalNet focused on the co-evolved residues and harnessed such information for prediction of metal binding. MetalNet learned features based on co-evolution information using an ML model and made the predictions after applying a graph filter. For protein citX, we remodeled its structure with a zinc-binding site as predicted by MetalNet, and the new structural model accurately matches the crystal structure that was solved later (Fig. 4g,h). Such data encouragingly suggested that MetalNet predictions can serve as a good complement to guide protein structural determination of metalloproteins by both experimental and computational methods.

For simplicity, MetalNet currently focused on metal-binding sites with CHED residues only. Intrinsically limited by the nature of sidechain co-evolution, MetalNet will not detect metal-binding sites with only one sidechain involved and therefore might make false negative predictions. Inevitably, this creates a bias against certain metal ions, such as Ca and Mg (Extended Data Fig. 4), whose chelating atoms often come from mainchain carbonyls, cofactors and waters. In addition, the graph filter might remove more potential sites with only two aminoacidic ligands. To alleviate the bias, we picked one pair from the ML model predictions when no pairs are left after filtering, which helped recover certain Ca or Mg sites. Moreover, because we mainly focused on the monomeric protein domains, certain metal-binding sites across domain or protein interfaces might not be predicted by MetalNet. In addition, MetalNet currently labels the co-evolution pairs with a general metal-binding tag and does not take into account different environments or oxidation states of metal ions. In the future, metal-type-specific prediction might be accomplished by training MetalNet with multiple tags and detailed sample classification.

Because MetalNet is critically dependent on the quality of co-evolution data derived from MSA, if the depth of MSA of a query protein is insufficient, the contact predictor may omit certain pairs of metal-binding residues. More sequencing data such as metagenome are needed for its application to eukaryotic systems, which was exemplified by our initial attempt on the Pfam families and human spliceosome. Moreover, because the prediction results of co-evolution-based MetalNet and structure-based MIB are only partially overlapping (Extended Data Fig. 6), it implies that these two methods may capture different features of metal-binding sites and it will therefore be inspiring to integrate additional sequence and structure-based features into MetalNet for improvement in future. Lastly, co-evolved networks should exist for functional sites not only with metals but also with other types of ligands or cofactors. With more experimental information obtained by chemoproteomic profiling[40–42] and structural models generated by AlphaFold[43] and RosettaFold[44], the pipeline of MetalNet should be readily extended for discovering binding sites for ligands and cofactors with larger and more complicated structures, and enzyme active sites with co-ordinated network architectures.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41589-022-01223-z.

References

1. Gladyshev, V. N. & Zhang, Y. Comparative genomics analysis of the metallomes. *Met. Ions Life Sci.* **12**, 529–580 (2013).
2. Waldron, K. J. & Robinson, N. J. How do bacterial cells ensure that metalloproteins get the correct metal? *Nat. Rev. Microbiol.* **7**, 25–35 (2009).
3. Yannone, S. M., Hartung, S., Menon, A. L., Adams, M. W. & Tainer, J. A. Metals in biology: defining metalloproteomes. *Curr. Opin. Biotechnol.* **23**, 89–95 (2012).
4. Waldron, K. J., Rutherford, J. C., Ford, D. & Robinson, N. J. Metalloproteins and metal sensing. *Nature* **460**, 823–830 (2009).
5. Cvetkovic, A. et al. Microbial metalloproteomes are largely uncharacterized. *Nature* **466**, 779–782 (2010).
6. Pace, N. J. & Weerapana, E. A competitive chemical-proteomic platform to identify zinc-binding cysteines. *ACS Chem. Biol.* **9**, 258–265 (2014).
7. Sevcenco, A. M. et al. Exploring the microbial metalloproteome using MIRAGE. *Metallomics* **3**, 1324–1330 (2011).
8. Andreini, C., Banci, L., Bertini, I. & Rosato, A. Counting the zinc-proteins encoded in the human genome. *J. Proteome Res.* **5**, 196–201 (2006).
9. Passerini, A., Punta, M., Ceroni, A., Rost, B. & Frasconi, P. Identifying cysteines and histidines in transition-metal-binding sites using support vector machines and neural networks. *Proteins Struct. Funct. Bioinf.* **65**, 305–316 (2006).
10. Passerini, A., Lippi, M. & Frasconi, P. MetalDetector v2.0: predicting the geometry of metal binding sites from protein sequence. *Nucleic Acids Res.* **39**, W288–W292 (2011).
11. Haberal, İ. & Oğul, H. Prediction of protein metal binding sites using deep neural networks. *Mol. Inf.* **38**, e1800169 (2019).
12. Babor, M., Gerzon, S., Raveh, B., Sobolev, V. & Edelman, M. Prediction of transition metal-binding sites from apoprotein structures. *Proteins* **70**, 208–217 (2008).
13. Lin, Y. F. et al. MIB: metal ion-binding site prediction and docking server. *J. Chem. Inf. Model.* **56**, 2287–2291 (2016).
14. Zhang, C., Freddolino, P. L. & Zhang, Y. COFACTOR: improved protein function prediction by combining structure, sequence and protein-protein interaction information. *Nucleic Acids Res.* **45**, W291–W299 (2017).
15. Gobel, U., Sander, C., Schneider, R. & Valencia, A. Correlated mutations and residue contacts in proteins. *Proteins* **18**, 309–317 (1994).
16. Shindyalov, I. N., Kolchanov, N. A. & Sander, C. Can three-dimensional contacts in protein structures be predicted by analysis of correlated mutations? *Protein Eng. Des. Select.* **7**, 349–358 (1994).
17. Martin, L. C., Gloor, G. B., Dunn, S. D. & Wahl, L. M. Using information theory to search for co-evolving residues in proteins. *Bioinformatics* **21**, 4116–4124 (2005).

18. Morcos, F. et al. Direct-coupling analysis of residue coevolution captures native contacts across many protein families. *Proc. Natl Acad. Sci. USA* **108**, E1293–E1301 (2011).
19. Balakrishnan, S., Kamisetty, H., Carbonell, J. G., Lee, S. I. & Langmead, C. J. Learning generative models for protein fold families. *Proteins* **79**, 1061–1078 (2011).
20. Jones, D. T., Buchan, D. W., Cozzetto, D. & Pontil, M. PSICOV: precise structural contact prediction using sparse inverse covariance estimation on large multiple sequence alignments. *Bioinformatics* **28**, 184–190 (2012).
21. Marks, D. S., Hopf, T. A. & Sander, C. Protein structure prediction from sequence variation. *Nat. Biotechnol.* **30**, 1072–1080 (2012).
22. Ovchinnikov, S. et al. Protein structure determination using metagenome sequence data. *Science* **355**, 294–298 (2017).
23. Xu, J. Distance-based protein folding powered by deep learning. *Proc. Natl Acad. Sci. USA* **116**, 16856–16865 (2019).
24. Ovchinnikov, S., Kamisetty, H. & Baker, D. Robust and accurate prediction of residue-residue interactions across protein interfaces using evolutionary information. *eLife* **3**, e02030 (2014).
25. Cong, Q., Anishchenko, I., Ovchinnikov, S. & Baker, D. Protein interaction networks revealed by proteome coevolution. *Science* **365**, 185–189 (2019).
26. Toth-Petroczy, A. et al. Structured states of disordered proteins from genomic sequences. *Cell* **167**, 158–170 (2016).
27. Chakrabarti, S. & Panchenko, A. R. Coevolution in defining the functional specificity. *Proteins* **75**, 231–240 (2009).
28. Kuipers, R. K. et al. Correlated mutation analyses on super-family alignments reveal functionally important residues. *Proteins* **76**, 608–616 (2009).
29. Chakrabarti, S. & Panchenko, A. R. Structural and functional roles of coevolved sites in proteins. *PLoS One* **5**, e8591 (2010).
30. Jeong, C. S. & Kim, D. Structure-based Markov random field model for representing evolutionary constraints on functional sites. *BMC Bioinf.* **17**, 99 (2016).
31. Wang, G. & Dunbrack, R. L. Jr. PISCES: a protein sequence culling server. *Bioinformatics* **19**, 1589–1591 (2003).
32. Anishchenko, I., Ovchinnikov, S., Kamisetty, H. & Baker, D. Origins of coevolution between residues distant in protein 3D structures. *Proc. Natl Acad. Sci. USA* **114**, 9122–9127 (2017).
33. Altschul, S. F., Gish, W., Miller, W., Myers, E. W. & Lipman, D. J. Basic local alignment search tool. *J. Mol. Biol.* **215**, 403–410 (1990).
34. Cariss, S. J. L. et al. YieJ (CbrC) mediates CreBC-dependent colicin E2 tolerance in Escherichia coli. *J. Bacteriol.* **192**, 3329–3336 (2010).
35. Schneider, K., Dimroth, P. & Bott, M. Biosynthesis of the prosthetic group of citrate lyase. *Biochemistry* **39**, 9438–9450 (2000).
36. Will, C. L. & Luhrmann, R. Spliceosome structure and function. *Csh Perspect. Biol.* **3**, a003707 (2011).
37. Charenton, C., Wilkinson, M. E. & Nagai, K. Mechanism of 5′ splice site transfer for human spliceosome activation. *Science* **364**, 362–367 (2019).
38. Zhang, X. F. et al. Structures of the human spliceosomes before and after release of the ligated exon. *Cell Res.* **29**, 274–285 (2019).
39. Zhang, C. X., Zheng, W., Mortuza, S. M., Li, Y. & Zhang, Y. DeepMSA: constructing deep multiple sequence alignment to improve contact prediction and fold-recognition for distant-homology proteins. *Bioinformatics* **36**, 2105–2112 (2020).
40. Piazza, I. et al. A map of protein-metabolite interactions reveals principles of chemical communication. *Cell* **172**, 358–372 (2018).
41. Zhuang, S., Li, Q., Cai, L., Wang, C. & Lei, X. Chemoproteomic profiling of bile acid interacting proteins. *ACS Cent. Sci.* **3**, 501–509 (2017).
42. Horning, B. D. et al. Chemical proteomic profiling of human methyltransferases. *J. Am. Chem. Soc.* **138**, 13335–13343 (2016).
43. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
44. Baek, M. et al. Accurate prediction of protein structures and interactions using a three-track neural network. *Science* **373**, 871–876 (2021).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2023

Methods

Benchmark set of known metal-binding proteins from PDB

We first downloaded a protein structure dataset used by Ivan et al. (ref. [32]). This dataset contains 9,846 protein chains that have corresponding structures deposited in PDB and whose MSAs were calculated by GREMLIN. For each PDB file, we used the Biopython[45] package to enumerate all the hetero-residues and collect metal ions, including ZN, CA, MG, CU, FE, NI, MN, CO, FE in SF4 (Fe_4S_4) and FES (Fe_2S_2). For each metal center, we then identified residues whose minimal distance between the sidechain hetero atom and the metal ion is less than 3 Å. Finally, we obtained 1,768 metalloproteins (Supplementary Dataset 1).

Training set for ML

We used hhmake and hhsearch to calculate a logarithmic 'E-value' for each protein in the benchmark, which can be used to evaluate the extent of sequence similarity between proteins in the dataset[46]. We then applied an exhaustive clustering based on the logarithmic 'E-value', which resulted in 771 nonoverlapping clusters. Next, we randomly picked one representative protein from each of these clusters and filtered off those proteins whose effective number (a measurement reflecting the quality of MSA, as calculated previously in ref. [22]) is less than ten. We finally obtained a set of 697 nonredundant metalloproteins.

Contact maps were calculated using MSA-transformer (https://www.biorxiv.org/content/10.1101/2021.02.12.430858v3, ESM-MSA-1b model from https://github.com/facebookresearch/esm). We used deepMSA to generate MSA and unifilt with option -diff 64 to shrink the MSA size. The script named contact_prediction.ipynb (from https://github.com/facebookresearch/esm) was used to predict contact probability. We tested different thresholds of the output contact signals by MSA transformer, and the highest F1 scores are within 0.58–0.72. We finally considered residue pair with predicted contact probability above 0.1 as contact. We also tried to use the contact signals calculated by another contact predictor, GREMLIN, and the highest F1 score of the whole method (with the graph filter) is around 0.69.

From the contact maps of these metalloproteins, we identified 2,346 co-evolved CHED pairs in which both residues are shown to bind metals in the structure. These 'metal-binding' co-evolved CHED pairs were used as the positive samples for training the ML model. We also obtained another set of 16,816 'nonmetal-binding' as negative samples in which neither of the residues is shown to bind metals in the structure.

For each of the 'metal-binding' or 'nonmetal-binding' co-evolved pairs, we calculated its 'pairwise frequency matrix' based on the residue–residue frequency profile observed in the homologous sequences from its original MSA. Each matrix is 21 × 21 by dimension, representing 20 types of proteinogenic amino acids and a gap ('–') in the aligned sequences. The matrix is ordered as CHDENSTKGQYLAVRIMFWP-. Matrices of the 'metal-binding' and 'nonmetal-binding' co-evolved pairs were fed as the positive and negative sample input, respectively, to train the ML model.

Training of the ML model

Before training, we shuffled the above-mentioned protein list randomly and used the fivefold cross-validation to split them into the training and validation sets. Ratio of the positive and negative samples (co-evolved pairs) that were used to train the ML model is adjusted to 1:1 to avoid bias.

We used AutoGluon (https://auto.gluon.ai/stable/index.html) to train a high-performance model. AutoGluon is an auto-machine-learning (AutoML) framework, which is developed by researchers in Amazon. Unlike existing AutoML frameworks that primarily focus on model/hyperparameter selection, AutoGluon-Tabular succeeds by ensembling multiple models, including XGBoost, Random Forest model, KNearestNeighbors model, neural network models and so on, and then stacking them in multiple layers. AutoGluon requires only a single line of Python to train highly accurate ML models on an unprocessed tabular dataset. The key parameters are shown as follows:

time_limit = 600,
metric = 'roc_auc',
predictor = TabularPredictor(label = label, eval_metric = metric,verbosity = 0).fit(train_data,
time_limit = time_limit,presets = 'best_quality').

Part of the predictor metadata are shown as below:

'random_state': 0, 'version': '0.2.0', 'time_fit_preprocessing': 0.3674032688140869, 'time_fit_training': 601.5672931671143, 'time_fit_total': 601.9346964359283, 'time_limit': 600, 'time_train_start': 1625713672.7824519, 'num_rows_train': 4,686, 'num_cols_train': 441, 'num_classes': 2, 'problem_type': 'binary', 'eval_metric': 'roc_auc', 'best_model': 'WeightedEnsemble_L3', 'best_model_score_val': 0.9380351899213791, 'best_model_stack_level': 3, 'num_models_trained': 25, 'num_bag_folds': 10, 'max_stack_level': 3, 'max_core_stack_level': 2

More details about the AutoGluon predictor can be found in Supplementary Table 4.

Complete metadata can be found in the source code with predictor.info().

Reported metrics for evaluating the performance of the ML model include accuracy, precision, recall, F1 score, AUC and MCC. All metrics were calculated using sklearn.metircs module.

We finally applied it to train the final model using all available samples, which was used for making predictions on other protein datasets as reported in the rest of the manuscript.

Identification of co-evolved network clusters

After predicting by ML whether each of the co-evolved CHED pairs is metal-binding, we used graph theory to detect if they can form co-evolved network clusters. This was done by separating out each unconnected subgraph in the predicted CHED co-evolution networks, and repeatedly filtering out those 'singleton' co-evolved pairs ('edge') in each subgraph. To do so, we first deleted all the residues ('nodes') with the connection number (degree) equal to one and their associated edges from the co-evolved CHED network, and then subtracted the degree of other nodes associated with these edges by one. The above procedure was repeated until all the nodes in the subgraph are connected by at least two associated edges. We tested different graph filter cutoffs, and finally set the cutoff as one according to the highest F1 score.

Construction of the co-evolved network pattern bank

For the 697 metalloproteins in the nonredundant training set mentioned above, we compiled the co-evolved network clusters and associated metal types to construct a 'co-evolution motif bank'. Because each 'motif' in the bank is a network, we used NetworkX[47] to record all necessary information of a network including the residue type of each node and the residue–residue distance in sequence for the two nodes connected by an edge. A total number of 239 networks were finally obtained.

When a new CHED co-evolution network is predicted by MetalNet, it can be queried against this co-evolution motif bank and the type of metal bound can be determined whether the network architecture is found to match any known metal-binding motif in the bank. NetworkX[47] is implemented to check whether the two networks are the same, based on the isomorphism, which takes into consideration the network geometry, residue type composition and residue distance in sequence ('F' for those with ≥3 amino acids away and 'N' for otherwise).

Comparison of MetalNet and MIB

We have done a 'prospective' analysis to compare MetalNet with another structure-based prediction method, MIB, in terms of predicting metal-binding residues from a common testing set. To avoid

unnecessary bias for either method, we collected all of the new metalloproteins in PDB released after 2016/08 because MetalNet used the protein structure dataset reported by Ivan et al.[32] as the benchmark (compiled on 2 August 2016) and MIB was published in 2016.

We downloaded the information table of protein entities from the PDB server (https://ftp.wwpdb.org/pub/pdb/derived_data/index/entries.idx) and only X-ray protein structures with ACCESSION DATE later than 08/02/16 and resolution of ≤2.0 Å were selected. We downloaded the reference sequence of the PDB structure from the PDB server (https://ftp.rcsb.org/pub/pdb/derived_data/pdb_seqres.txt.gz) and selected protein chains with sequence lengths between 40 and 600. We used the online CD-HIT server[48] (http://weizhong-lab.ucsd.edu/cdhit_suite/cgi-bin/index.cgi?cmd=cd-hit) to cluster these peptides according to their sequence similarity with the default parameter. At this stage, we obtained 4,561 representative protein chains.

Next, we used colabfold[49] to generate MSA for these proteins and hhmake from hhsuite3 to generate HMM based on their MSA[46]. We used hhsearch to calculate a logarithmic 'E-value' for proteins and our trainset to remove proteins that have homologs in the trainset of MetalNet[46]. We then used hhsearch to evaluate the extent of sequence similarity between the remaining 373 proteins and applied an exhaustive clustering, which resulted in 342 nonoverlapping clusters. The selection criteria for the representative protein of each protein cluster is the protein with the highest NEFF, which is obtained from the HMM file. In the end, a total of 60 metalloproteins were obtained.

Metal-binding residues in the PDB structure files are extracted as mentioned above. In total, there are 227 metal-binding CHED residues in these 60 metalloproteins. Considering that MetalNet is a co-evolution-dependent method and will not detect metal-binding sites with only one sidechain, we also removed such cases from the testing test, resulting in a total of 214 metal-binding residues in the remaining 47 metalloproteins. We compared the performance of MetalNet and MIB in predicting specific metal ions, including Zn, Mg/Ca and others.

We used the reference sequence recorded in the PDB of related protein as the input of MetalNet and utilized sequence alignment to assign correct residue numbers if necessary. The PDB ID (code) was used as the input of MIB server (http://bioinfo.cmu.edu.tw/MIB/structure/) for prediction. Precision, recall and F1 score were calculated on the residue level. Venn diagram was generated using BioVenn[50].

Dataset of four prokaryotic species

The four species include *E. coli* (strain K12; 83333), *B. subtilis* (strain 168; 224308), *S. solfataricus* (strain ATCC 35092/DSM 1617/JCM 11322/P2; 273057) and *H. salinarum* (strain ATCC 700922/JCM 11081/NRC-1; 64091). The MSA of each proteome was downloaded from GREMLIN (https://gremlin2.bakerlab.org/db/*/fasta/, with * standing for the abbreviation of the four species. ECOLI for *E. coli*, BACSU for *B. subtilis*, SULSO for *S. solfataricus* and HALSA for *H. salinarum*), which include 4,305 proteins from *E. coli*, 4,093 from *B. subtilis*, 2,938 from *S. solfataricus*, and 2,421 from *H. salinarum*. We finally obtained 10,670 effective MSAs. We used MSA transformer to calculate contact signals and successfully obtained 10,259 contact maps.

UniProt annotation

For a given species, we downloaded from UniProt (https://ebi10.uniprot.org) profiles (date: 16 August 2021) of its whole proteome and extracted protein entries containing the 'FT METAL' annotation and the detailed information of the liganding residues. The numbers of the annotated metalloproteins by this measure are 950 in *E. coli*, 651 in *B. subtilis*, 149 in *S. solfataricus* and 215 in *H. salinarum*. Such information was used to cross-validate predictions by MetalNet.

Direct structural evidence for MetalNet predictions

To seek direct structural evidence that can support the MetalNet predictions, we first searched for their homologs in the benchmark. We used hhmake to obtain the hhm alignment for the query protein and hhsearch to calculate the log (*E* value) between the query protein and each protein in the benchmark. Any protein with a calculated log (*E* value) of more than 20 is considered as a hit. If there is no homolog found in the benchmark, we then ran BLASTP (version 2.7.1+) against the PDB database (16 January 2018 release, ftp://ftp.ncbi.nlm.nih.gov/blast/db/pdbaa.tar.gz) with all the parameters set as default (including the *E* value = 10) to get all remote homologs.

The Biopython package[45] was used to align the query and homolog sequences. If all the predicted metal-binding residues by MetalNet are mapped to gaps or non-CHED residues in the homolog sequences, the protein was removed from further analysis. If a matching homolog is found, we next checked if the structure is metal-binding, and if so, we calculated the minimal distance between each of the mapped residues to the metal ion and assigned the residue as 'liganding' if the distance is less than 4 Å. The predicted metal-binding site by MetalNet is deemed to have direct structural evidence only if more than one residue in the site is found to be 'liganding' in the homolog structure. In the case when multiple homolog structures are found, the one that is mapped with the largest number of 'liganding' residues and the lowest *E* value will be chosen.

Indirect evidence to support MetalNet predictions

For every MetalNet prediction that lacks support from direct structural evidence, we searched for indirect but relevant evidence from various sources to further increase confidence.

1. We searched, for each prediction, whether the predicted metal-liganding residues are annotated in the UniProt record with the explicit FT METAL annotation.
2. We searched the Pfam database (http://pfam.xfam.org/) to retrieve the protein family information for the predicted protein. If a keyword ('metal', 'zinc', 'calcium', 'magnesium', 'copper', 'iron', 'nickel', 'manganese', 'iron–sulfur/sulfur', 'cobalt') is mentioned in the name or abstract of the protein family, the queried protein is considered to be functionally related to metal binding.
3. We searched the Gene Ontology (GO) database (https://www.ebi.ac.uk/QuickGO/) for the GO annotations including keywords in the function term.

Metagenome-pfam dataset

The dataset and structural model were previously reported in ref. [22]. We downloaded related MSAs from https://gremlin2.bakerlab.org/db/UNI/fasta/ and then calculated co-evolution signals using MSA transformer as in the prokaryotic species dataset. The mean distance between predicted pairs on structural model was calculated by Biopython with homemade script.

Human spliceosome dataset

From four high-resolution (≤3 Å) cryoEM structures of the human spliceosome in PDB (codes: 6ID0, 6ID1, 6ICZ and 6QW6), we collected a total of 57 subunit proteins and extracted their sequences from the associated UniProt records. These proteins were further divided into 77 domains using Pfam information and their MSAs were generated by deepMSA with the default setting. The co-evolution signals were then calculated by MSA transformer.

Zinc-binding sites on the structures include:
Site 1: 6ID1_N ZN201 ZN202 ZN203
Site 2: 6ID1_O ZN501 ZN502 ZN503
Site 3: 6ID1_U ZN900
Site 4: 6ICZ_Z ZN601
Site 5: 6QW6_U ZN600

Molecular cloning

The full-length citX was constructed into the pET28a vector with an N-terminal his-tag by amplification from the corresponding

chromosomal DNA using the primers that are listed in Supplementary Table 5 with the tab names 'citX-F and citX-R'.

The NdeI and XhoI sites were used in cloning. The variants were generated by QuikChange site-directed mutagenesis using the primers that are listed in Supplementary Table 5 with the tab names 'citX-(variant)-F and citX-(variant)-R'.

All vectors were verified by sequencing and subsequently transformed into the *E. coli* strain BL21 (DE3).

The full-length cbrC was constructed into the pMAL-C2X vector with a C-terminal his-tag and an N-terminal MBP-tag by amplification from the corresponding chromosomal DNA using the primers that are listed in Supplementary Table 5 under the section 'cbrC-F and cbrC-R'.

The BamHI and HindIII sites were used in cloning. The variants were generated by QuikChange site-directed mutagenesis using the primers that are listed in Supplementary Table 5 under the section 'cbrC-(variant)-F and cbrC-(variant)-R'.

The full-length desor_0198 was constructed by bridge-PCR using the primers that are listed in Supplementary Table 5 under the section 'desor_0198-F0(1/2/3) and desor_0198-R0(1/2/3)'.

The NdeI and XhoI sites were used to construct into the pET28a vector with an N-terminal his-tag using the primers that are listed in Supplementary Table 5 under the section 'desor_0198-enzyme-F and desor_0198-enzyme-R'.

The full-length SVEN_5263 was artificially synthesized and then constructed into the pD2P_1.06e vector with a C-terminal his-tag and an eGFP tag (from Kangma (Shanghai) Biotechnology Co, Ltd). The primers that are listed in Supplementary Table 5 under the section 'SVEN_5263-F and SVEN_5263-R' were further used by PCR amplification to obtain the plasmid without the eGFP sequence for ICP-MS analysis.

Protein expression and purification

To obtain protein citX for crystallization, cells were cultured in M9 minimal medium at 37 °C until reaching an OD600 = 0.4–0.6, and then added with 100 mg each of phenylalanine, threonine, lysine-HCL and 50 mg each of selenomethionine, leucine, valine and isoleucine. For protein samples used in circular dichroism, thermal shift assay, gel analysis and ICP-MS bacteria were cultured directly in LB medium at 37 °C until OD600 = 0.6–0.8. The protein overexpression was induced by the addition of isopropyl β-D-1-thiogalactopyranoside (IPTG) to a final concentration of 0.5 mM.

After IPTG induction at 18 °C for 18 h, bacterial cells were collected by centrifuging at 3,000 g for 30 min. The citX-overexpressed bacteria were resuspended in 25 mM Tris, pH 8.5, 1.0 M NaCl and 1 mM phenylmethanesulfonyl fluoride (PMSF), and the buffer containing 25 mM Tris, pH 8.0, 0.5 M NaCl and 1 mM PMSF was used for cbrC-overexpressed bacteria.

Next, the cells were lysed by sonication and centrifuged at 25,000 g for 90 min. For preliminary purification of the target protein, the supernatant fraction was loaded onto a Ni-chelating column (GE Healthcare, USA), and target proteins were eluted by imidazole gradient at a concentration of 150–500 mM.

Protein samples of citX were mixed with thrombin (~1 U/1 mg) in 25 mM Tris, pH 8.5 and 100 mM NaCl for 36 h before a final purification by size-exclusion chromatography (Superdex 75, GE Healthcare) in 25 mM Tris, pH 8.0 and 100 mM NaCl.

For protein samples of cbrC, the MBPSep dextrin 6FF chromatography column (Yeasen) was used for further purification, and the target protein was eluted by 25 mM Tris, pH 8.0, 100 mM NaCl and 20 mM maltose.

Circular dichroism

Purified citX samples were diluted to 0.3 mg ml^{-1} in PBS (pH 8.5) in a 0.1-cm cuvette. The samples were treated with different concentrations of EDTA (2.5 mM, 5.0 mM, 10 mM and 20 mM) and incubated at 4 °C for 30 min before scanning. Circular dichroism (CD) analysis was performed using a Bio-Logic MOS-500 CD spectrometer at scanning wavelengths of 250–190 nm.

Thermal shift assay

Purified citX samples (0.2 μg $μl^{-1}$) were treated with 0.94 mM EDTA or PBS buffer alone. The mixture was incubated for 30 min at 25 °C, divided into six aliquots and transferred into 0.2-ml PCR tubes. All of the EDTA or PBS-treated samples were heated in parallel for 3 min to their specified temperatures, respectively. The samples were subsequently centrifuged at 20,000*g* for 20 min at 4 °C and the supernatant was collected for SDS-PAGE analysis. The gel was stained by coomassie blue and the intensity of the bands was quantified by ImageJ. GraphPad was utilized to fit the curve and the temperature when half of the protein was unfolded is identified as the melting temperature (Tm).

Gel analysis of wild-type proteins and variants

After culturing at 18 °C for 18 h, equal amounts of bacteria were normalized by OD600 and pelleted by centrifugation at 4,000*g* for 10 min. The pellets were added with 50 μl of 2X gel loading buffer and boiled for 10 min at 95 °C. The resulting samples were used for gel analysis of the whole proteome content.

To obtain the soluble fraction, the bacteria pellets were resuspended in the buffer of 25 mM Tris, pH 8.5, 1.0 M NaCl and lysed by sonication. The samples were centrifugated and the protein concentration of the supernatant was determined by the BCA protein assay (Pierce BCA Protein Assay Kit, Thermo Fisher Scientific). The samples were diluted to 1 mg ml^{-1} and added with 5X gel loading buffer followed by boiling for 10 min at 95 °C. The samples were separated by 10% SDS-PAGE, stained by co-omassie blue and visualized by a Bio-Rad ChemiDoc MP Imaging System.

ICP-MS

The protein samples (100 μl) were added with 500 μl nitric acid and 200 μl hydrogen peroxide and then boiled overnight to decompose the organic compounds. The samples were diluted by 50 folds using water and the metal contents (zinc, iron and copper for citX and zinc for cbrC) were analyzed by ICP-MS using an Elan DRC-e (PerkinElmer) spectrometer. The protein concentrations were determined by Nanodrop and BCA kit.

Modeling the structure of citX with metal binding

The loops with the four predicted metal-binding residues of the apo citX model were deleted and the remaining structure was taken as a template for modeling by the RosettaCM protocol[51]. Two types of constraints were implemented—the distance between the C atom of each liganding residue and the metal ion in the centroid stage, and the distance and angle constraints were in the full-atom stage. All the geometry parameters were used as previously reported[52].

Crystallization of citX

Purified citX samples were concentrated to approximately 8.0 mg ml^{-1} in 25 mM Tris pH 8.0 and 100 mM NaCl. The following crystallization screens were used to get crystals: Crystal Screen (Hampton Research), Index (Hampton Research), ProPlex (Molecular Dimensions) and Wizard classic1–4 (Rigaku). The optimal conditions for crystallization of the different designs were found as follows: 0.15 M $(NH_4)_2SO_4$, 0.1 M MES, pH 5.5, 20% (wt/vol) polyethylene glycol 4,000 with an additional 8% (vol/vol) (±)-1, 3-butanediol. Crystals were obtained after 1–7 d with the drops consisting of a 1:1 mixture of protein solution and reservoir solution.

X-ray data collection and structure determination

The protein crystals were picked by cryo-loops and placed in the corresponding reservoir solution, containing 20% (vol/vol) glycerol

and flash-frozen in liquid nitrogen. The X-ray data were collected at Beamline 17U1 and BL19U of Shanghai Synchrotron Radiation Facility at the Se peak wavelength 0.9792 Å with Eiger 16 M detector, and the samples were kept at 100 K during data collection. The initial model was determined by single-wavelength anomalous diffraction phasing using the anomalous signal from Se atoms with SHELX[53] using the peak data and finally refined to a resolution of 2.5 Å. Model building and refinement were carried out iteratively using Phenix[54] and Coot[55]. Then another higher-resolution structure was determined by molecular replacement with Phenix AutoMR using the initial model as the search model and finally refined to a resolution of 1.7 Å. Final structures were validated with the wwPDB server. The data processing and refinement statistics are summarized in Supplementary Table 3 (from coot).

Molecular dynamics

Molecular dynamics (MD) simulations were performed to calculate root mean square fluctuation (RMSF) for citX (PDB code: 7DCN) and its variants. We utilized the GROMACS 2019.3 package and its analytical tools[56]. 'Mutagenesis' tool within PyMOL[57] was used for constructing the citX variants. Each system was solvated in a cubic box filled with TIP3P water model and neutralized by Na^+ ions. Each system was then minimized and two equilibration steps were set up under constant temperate (NVT) and constant pressure (NPT) for 0.1 ns, respectively. For production runs, the temperature was coupled with V-rescale method and the pressure was coupled with the Parrinello–Rahman method[58]. The Particle Mesh Ewald method was used for calculation of long-range electrostatics[59]. Each system was simulated for 40 ns and the last 10 ns trajectory was used for RMSF calculation.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The original protein structure dataset containing 9,846 protein sequences and their MSA can be downloaded from https://doi.org/10.1073/pnas.1702664114. Co-evolved pairs used in model training can be found in Supplementary Dataset. The proteins and related MSA of prokaryotic species can be downloaded from https://gremlin2.bakerlab.org/db/{species}/fasta/. The Metagenome-pfam MSA and structural model can be downloaded from https://gremlin2.bakerlab.org/db/UNI/. The human spliceosome dataset can be found in Supplementary Dataset. PDB codes 6ID0, 6ID1, 6ICZ and 6QW6 were used to construct human spliceosome dataset. We downloaded the information table of protein entities from the PDB server (https://ftp.wwpdb.org/pub/pdb/derived_data/index/entries.idx) to construct unbiased dataset when comparing methods. UniProt (https://ebi10.uniprot.org) profiles (date: 16 August 2021), Gene Ontology database (https://www.ebi.ac.uk/QuickGO/) and Pfam database (http://pfam.xfam.org/) were used in the analysis. The pdbaa database (16 January 2018 release, ftp://ftp.ncbi.nlm.nih.gov/blast/db/pdbaa.tar.gz) is used in BLASTP. The structures of citX reported in this paper have been deposited in PDB with the accession numbers 7DCM (determined by single-wavelength anomalous diffraction) and 7DCN (determined by molecular replacement). Source data are provided with this paper.

Code availability

Our code is available as open source at https://github.com/wangchulab/MetalNet.

References

45. Cock, P. J. A. et al. Biopython: freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics* **25**, 1422–1423 (2009).
46. Steinegger, M. et al. HH-suite3 for fast remote homology detection and deep protein annotation. *BMC Bioinform.* **20**, 473 (2019).
47. Varoquaux, G., Vaught, T., & Millman, J. (eds.). Exploring network structure, dynamics, and function using networkX. In *Proceedings of the 7th Python in Science Conference* 11–15 (SciPy, 2008).
48. Huang, Y., Niu, B. F., Gao, Y., Fu, L. M. & Li, W. Z. CD-HIT suite: a web server for clustering and comparing biological sequences. *Bioinformatics* **26**, 680–682 (2010).
49. Mirdita, M. et al. ColabFold: making protein folding accessible to all. *Nat. Methods* **19**, 679–682 (2022).
50. Hulsen, T., de Vlieg, J. & Alkema, W. BioVenn—a web application for the comparison and visualization of biological lists using area-proportional Venn diagrams. *BMC Genom.* **9**, 488 (2008).
51. Song, Y. F. et al. High-resolution comparative modeling with RosettaCM. *Structure* **21**, 1735–1742 (2013).
52. Wang, C., Vernon, R., Lange, O., Tyka, M. & Baker, D. Prediction of structures of zinc-binding proteins through explicit modeling of metal coordination geometry. *Protein Sci.* **19**, 494–506 (2010).
53. Sheldrick, G. M. Experimental phasing with SHELXC/D/E: combining chain tracing with density modification. *Acta Crystallogr. D Biol. Crystallogr.* **66**, 479–485 (2010).
54. Adams, P. D. et al. PHENIX: building new software for automated crystallographic structure determination. *Acta Crystallogr. D Biol. Crystallogr.* **58**, 1948–1954 (2002).
55. Emsley, P. & Cowtan, K. Coot: model-building tools for molecular graphics. *Acta Crystallogr. D Biol. Crystallogr.* **60**, 2126–2132 (2004).
56. Abraham, M. J. et al. GROMACS: high-performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* **1**, 19–25 (2015).
57. PyMOL. *The PyMOL Molecular Graphics System, Version 2.4* (Schrodinger Inc., 2015).
58. Bussi, G., Donadio, D. & Parrinello, M. Canonical sampling through velocity rescaling. *J. Chem. Phys.* **126**, 014101 (2007).
59. Wang, H., Dommert, F. & Holm, C. Optimizing working parameters of the smooth particle mesh Ewald algorithm in terms of accuracy and efficiency. *J. Chem. Phys.* **133**, 034117 (2010).

Acknowledgements

We thank H. Tang in Chu Wang's lab and the Institute of Geographic Sciences and Natural Resources Research, CAS for the help with ICP measurements. We thank National Center for Protein Sciences at Peking University, Beijing for the help with Circular Dichroism measurements, and the staff of the Shanghai Synchrotron Radiation Facility and KEK Photon Factory for assistance with X-ray data collection. Funding: C.W. was supported by the National Natural Science Foundation of China (grants 21925701, 91953109 and 92153301).

Author contributions

H.W. and C.W. conceived the project; Y.C., H.W. and Y.L. performed computational analysis with the help of C.S. and S.O.; H.X. purified citX and solved the structure under the guidance of X.S.; B.M., X.C., X.W. and X.Z. contributed to biochemical verification of zinc binding in citX and other proteins; and Y.C, H.W., Y.L. and C.W. analyzed the data and wrote the manuscript with inputs from all authors.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41589-022-01223-z.

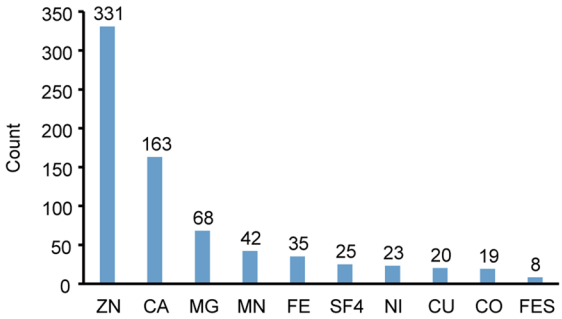
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41589-022-01223-z.

Correspondence and requests for materials should be addressed to Yuan Liu, Xiao-Dong Su or Chu Wang.

Peer review information *Nature Chemical Biology* thanks Kevin Yang, Rosalin Bonetta Valentino and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

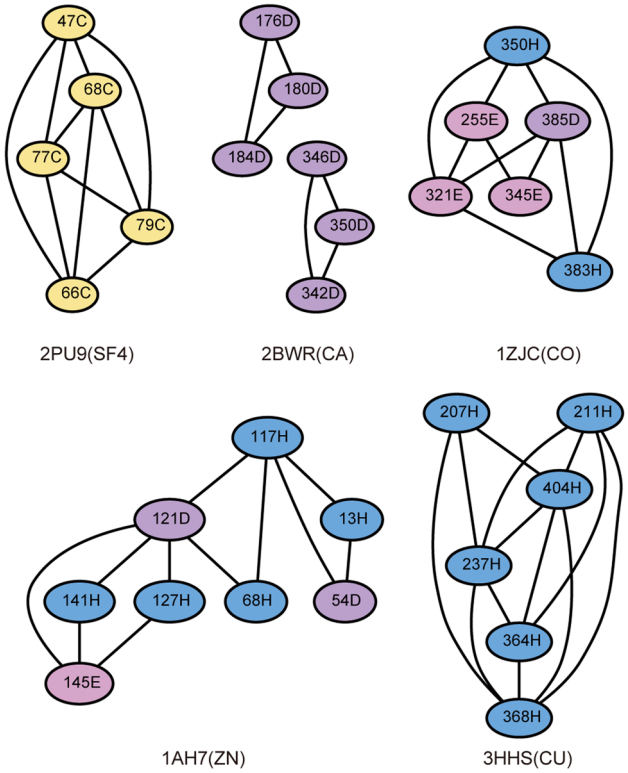
Reprints and permissions information is available at www.nature.com/reprints.

https://doi.org/10.1038/s41589-022-01223-z



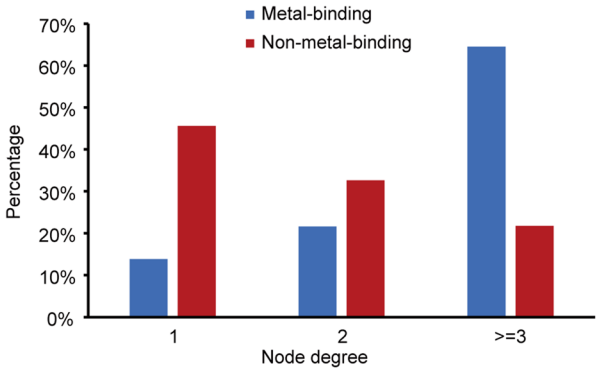
Extended Data Fig. 1 | Distribution of the metal types of the metalloproteins in the training set. ZN: zinc; CA: calcium; MG: magnesium; MN: manganese; FE: iron; SF4: [FE4-S4] iron-sulfur clusters; NI: nickel; CU: copper; CO: cobalt; FES: [FE2-S2] iron-sulfur clusters.

https://doi.org/10.1038/s41589-022-01223-z



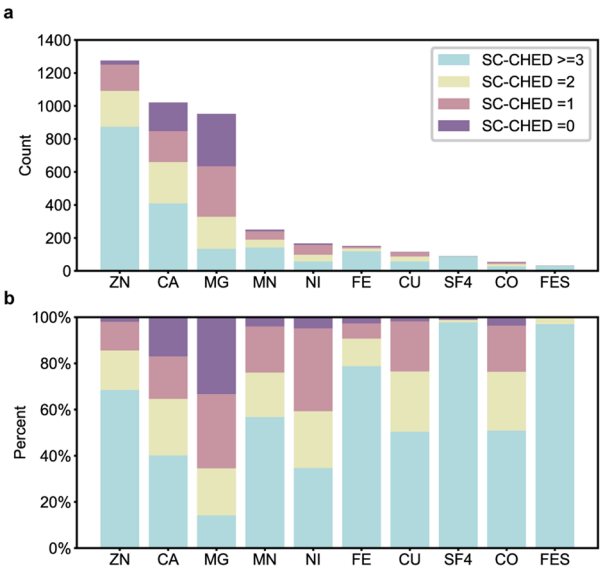
Extended Data Fig. 2 | Examples of the coevolved CHED network clusters detected in the PDB data. Metal-binding residues in these sites have the tendency to coevolve with each other and cluster together in a high-order coevolution network. The PDB ID and type of the metal ion bound are listed below the corresponding coevolved network.

https://doi.org/10.1038/s41589-022-01223-z



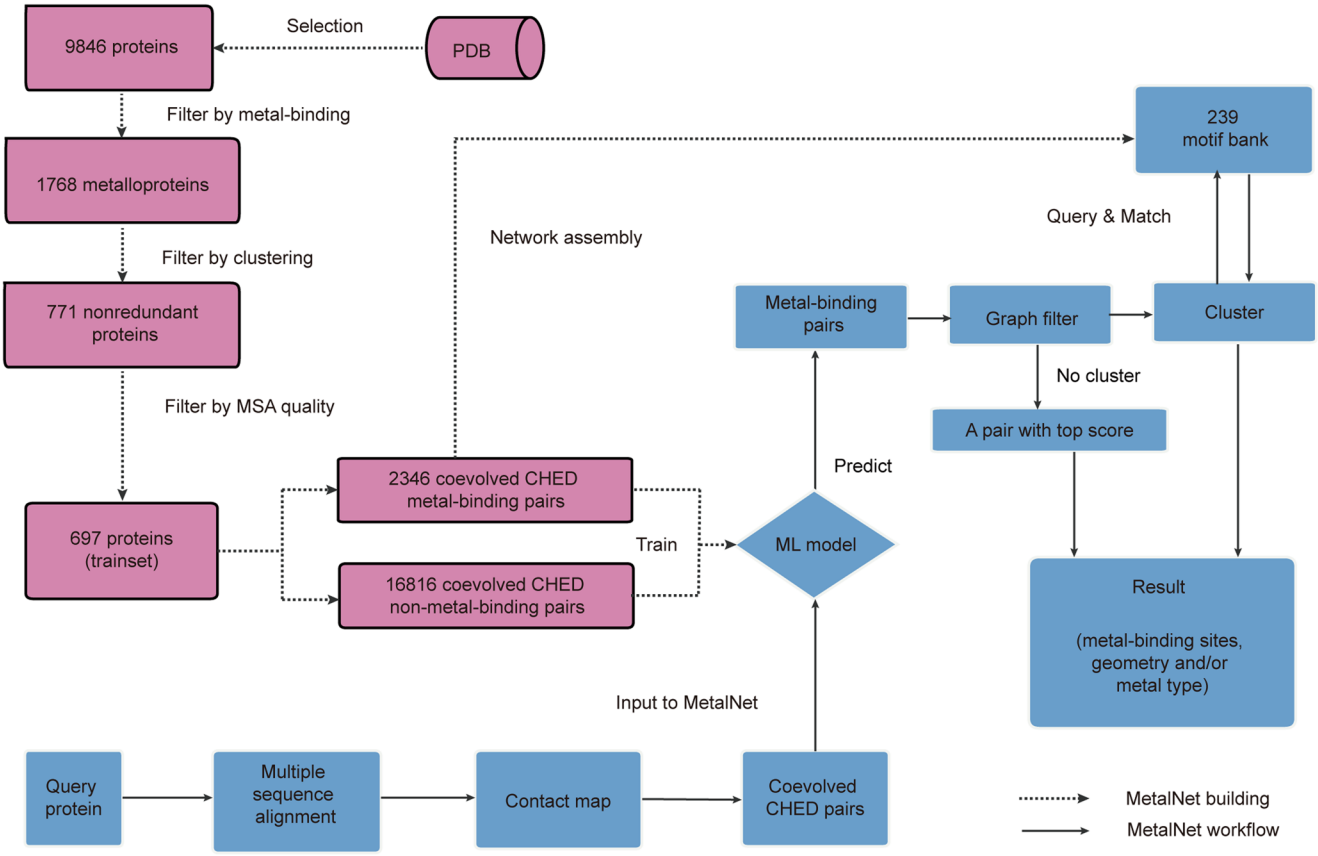
Extended Data Fig. 3 | Distribution of the number of coevolved connections ('node degree') for each residue among all the coevolved CHED pairs in the training set. The CHED residues involved in metal binding (blue) have more node degrees on average than those non-metal-binding residues in the coevolved networks (red).

https://doi.org/10.1038/s41589-022-01223-z



Extended Data Fig. 4 | Composition of metal-chelating CHED sidechains within the coordination sphere around each specific metal ion. The statistic is calculated based on metalloprotein structures in the benchmark. Figure a and b show the absolute count and normalized percentage, respectively. ZN: zinc; CA: calcium; MG: magnesium; MN: manganese; NI: nickel; FE: iron; CU: copper; SF4: [FE4-S4] iron-sulfur clusters; CO: cobalt; FES: [FE2-S2] iron-sulfur clusters.

https://doi.org/10.1038/s41589-022-01223-z



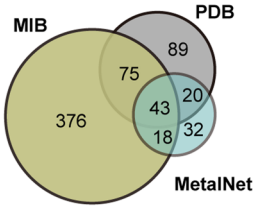
Extended Data Fig. 5 | Overall workflow of MetalNet. Starting with the information from coevolution analysis, MetalNet uses a machine learning (ML)-based classifier that has been trained by a benchmark of known metalloproteins to predict whether an individual CHED coevolved residue pair is metal-binding or not. It then employs a graph-based approach to identify high-order coevolution network clusters formed by these coevolved CHED pairs to generate reliable predictions of metal-binding sites in the query protein. In certain cases when the coevolution network topology can be matched to that of known metal-binding sites, the method can also infer the information on the type of metal bound in the predicted site. The method does not use any sequence (1D) or structural (3D) homology information to make predictions.

https://doi.org/10.1038/s41589-022-01223-z

a

Full Dataset with 60 metalloproteins that were released in PDB after 2016/08

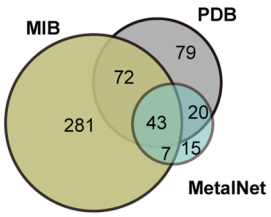
	Residue count	Precision		Recall		F1-score	
		MIB	MetalNet	MIB	MetalNet	MIB	MetalNet
All sites	227	23.0%	55.8%	52.0%	27.8%	0.319	0.371
Zn	120	37.5%	68.2%	65.0%	37.5%	0.476	0.484
Mg/Ca	67	9.0%	22.2%	28.4%	11.9%	0.137	0.155
others	40	22.6%	90.9%	52.5%	25.0%	0.316	0.392



b

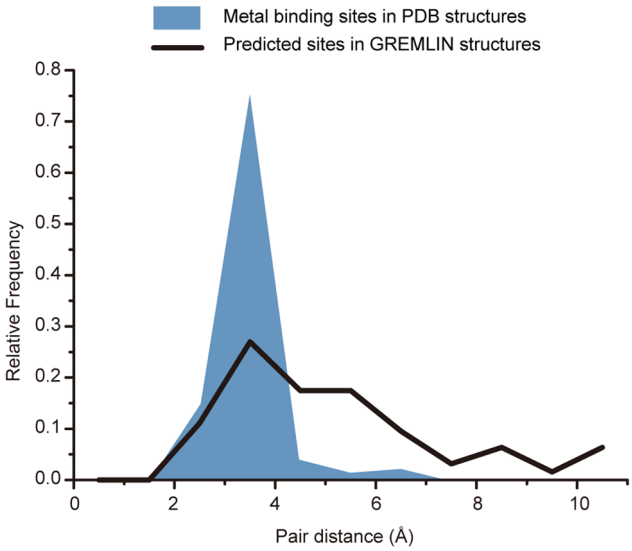
Trimmed dataset after cases with only one sidechain as the chelating group removed

	Residue count	Precision		Recall		F1-score	
		MIB	MetalNet	MIB	MetalNet	MIB	MetalNet
All sites	214	28.5%	74.1%	53.7%	29.4%	0.373	0.421
Zn	119	38.2%	75.0%	65.5%	37.8%	0.483	0.503
Mg/Ca	56	14.4%	57.1%	30.4%	14.3%	0.195	0.229
others	39	24.7%	90.9%	51.3%	25.6%	0.333	0.400



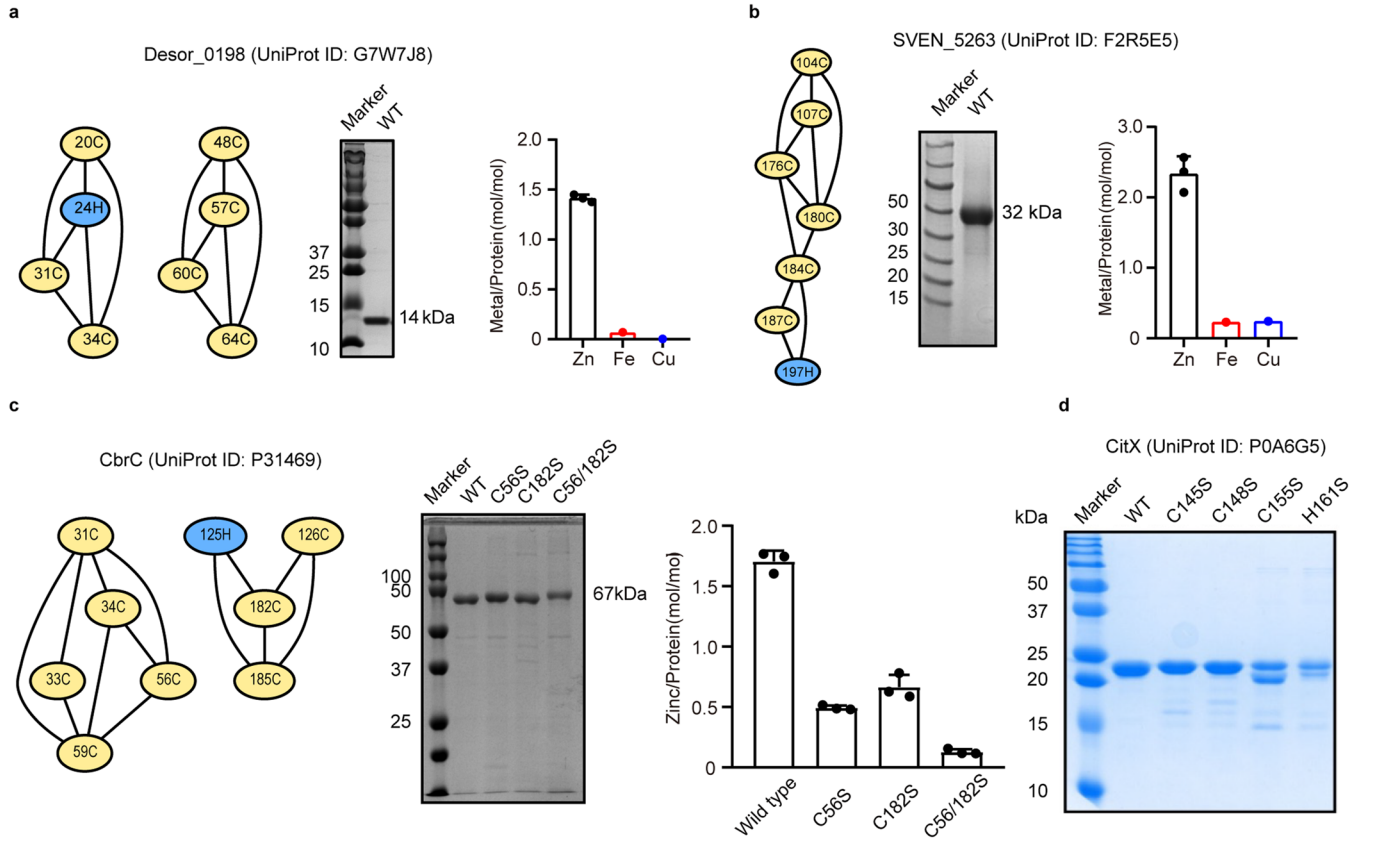
Extended Data Fig. 6 | Comparation of the performance between MetalNet and MIB. a, Evaluation of the performance of MetalNet and MIB on a 'prospective' metalloprotein dataset that were deposited in PDB after 2016/08. b, Evaluation of the performance of MetalNet and MIB on this prospective metalloprotein dataset after proteins with one single sidechain as the liganding group were removed. In a and b, the calculated precision, recall and F1-scores for MetalNet and MIB were shown on the left in the table format for all metal-binding sites ('All sites'), Zn-specific sites ('Zn'), Mg/Ca-specific sites ('Mg/Ca') and the remaining sites ('others'). The number of correct and incorrect predictions made by MetalNet and MIB were shown on the right in the venn diagram format with the 'prospective' metalloprotein dataset ('PDB') as the reference (the dataset can be found in Supplementary Dataset 3).

https://doi.org/10.1038/s41589-022-01223-z



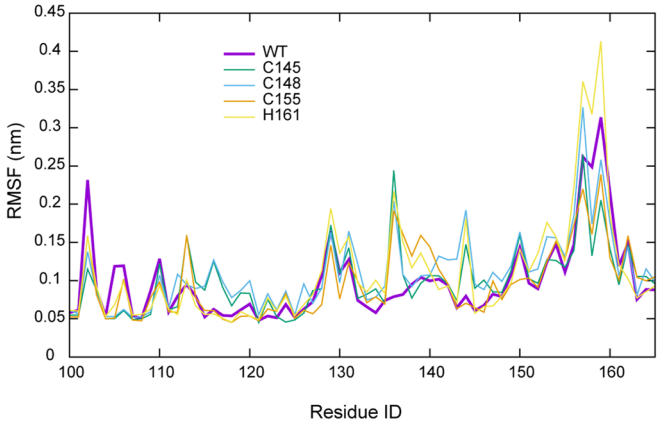
Extended Data Fig. 7 | Distribution of the minimal distances between the metal-binding residue pairs predicted by MetalNet. The blue block shows the distribution of residue-residue pairwise distances for metal-binding residues from known metalloproteins in the PDB database. Black solid lines show the distribution of mean distances in GREMLIN models for the metal-binding coevolved sites predicted by MetalNet. The distributions generally agree well with each other until 4.5 Å, suggesting that a large portion of MetalNet predictions indeed have metal-binding residues in proximity with each other. Since GREMLIN does not take metal-binding into consideration during structural modeling, the disagreement after 4.5 Å may come from either false positive prediction by MetalNet, or some errors in structural modeling by GREMLIN.

https://doi.org/10.1038/s41589-022-01223-z



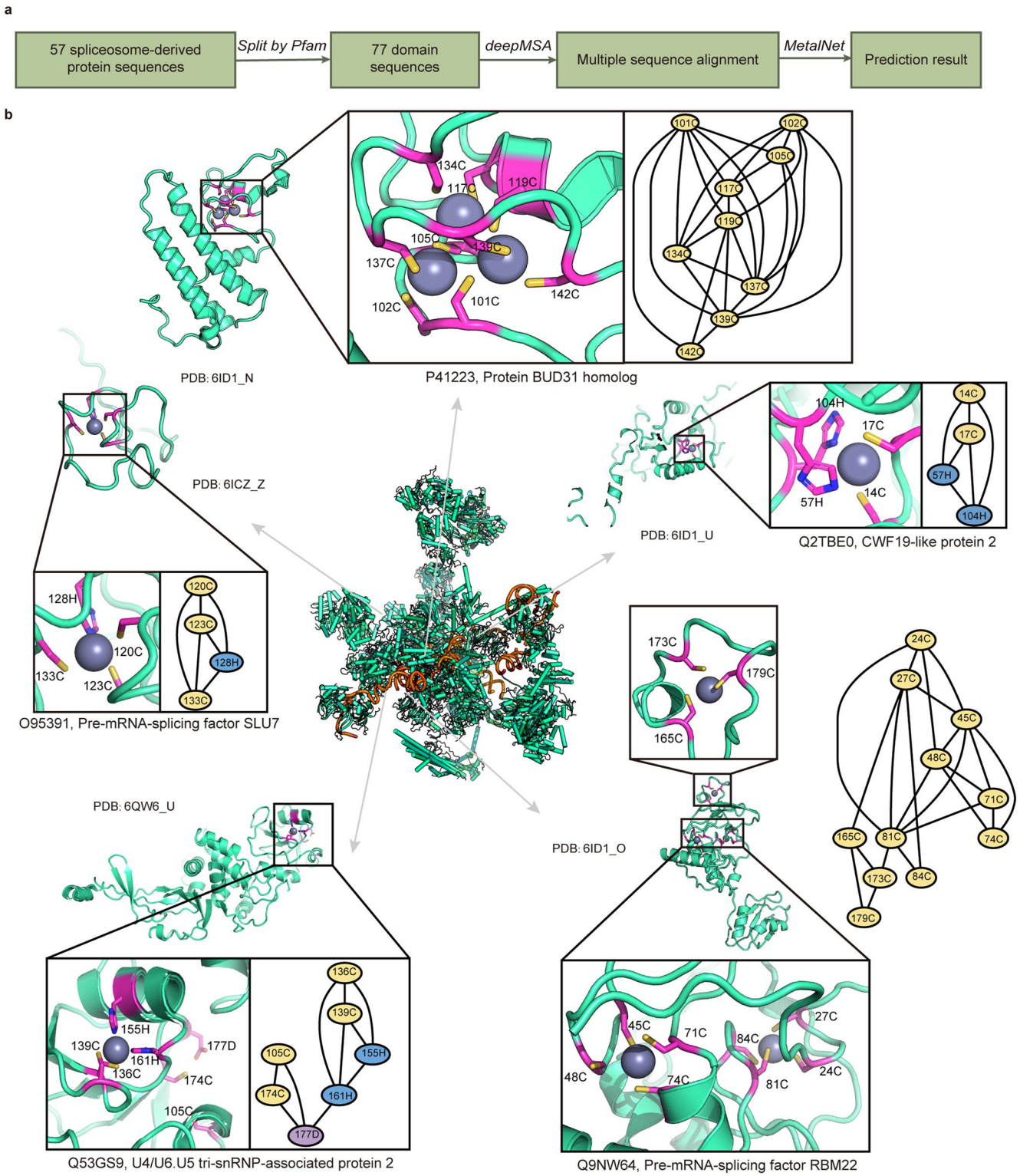
Extended Data Fig. 8 | Validation of metal binding in purified Desor_0198 (a), SVEN_5263 (b), CbrC (c) and CitX (d) by ICP-MS. For a and b, the coevolved metal-binding cluster predicted by MetalNet are shown on the left, SDS-PAGE of the purified wild-type protein in the middle and ICP-MS measurement of metal binding of the purified protein are shown on the right. Both proteins are validated with zinc-binding activity. c, The coevolved metal-binding cluster predicted by MetalNet are shown on the left, SDS-PAGE of the purified wild-type protein, two single mutants (C56S or C182S) and the double mutant (C56S&C182S) in the middle and ICP-MS analysis of zinc binding in the purified wild-type cbrC and mutants on the right. Partial zinc binding was retained in each of the single mutant whereas metal binding activity was completely abolished in the double mutant. In a, b, and c, error bars mean and s.d (n = 3 biologically independent samples). ICP-MS analysis of Fe and Cu binding is measured only once. d, SDS-PAGE of the purified wild-type citX as well as four single mutants of the predicted metal-binding residues by MetalNet (C145S, C148S, C155S and H161S). ICP-MS analysis of citX is shown in Fig. 4d. The experiment was repeated twice independently with similar results.

https://doi.org/10.1038/s41589-022-01223-z



Extended Data Fig. 9 | Molecular dynamic (MD) simulations of citX and its mutants. MD simulations were performed to calculate root mean square fluctuation (RMSF) for citX (PDB ID: 7DCN) and its mutants (C145S, C148S, C155S and H161, metal-binding residues predicted by MetalNet) using GROMACS. The results suggested that all mutants show greater conformational fluctuations at the binding site.

https://doi.org/10.1038/s41589-022-01223-z



Extended Data Fig. 10 | Prediction of metal-binding sites in the human spliceosome. a, Scheme of predicting metal-binding sites in the human spliceosome using coevolution obtained by deepMSA and MSA transformer. b, Highlight of the zinc-binding sites in the human spliceosome predicted by MetalNet that match well with the experimental structures in PDB. The experimental structures of the corresponding metalloproteins subunits are shown in cartoon on the left and the predicted coevolution networks corresponding to the metal-binding sites are shown on the right. Metal-chelating residues are shown in sticks and zinc ions are shown in spheres.

nature portfolio

Corresponding author(s): Chu Wang

Last updated by author(s): Oct 26, 2022

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used.
Data analysis	Following tools and packages were used: GREMLIN(v2.01), BioPython(v1.70), hhsuite(v2.0.16), MSA-transformer [https://www.biorxiv.org/content/10.1101/2021.02.12.430858v3, ESM-MSA-1b(v0.3.1) model from https://github.com/facebookresearch/esm], deepMSA[https://zhanggroup.org/DeepMSA/package.zip], AutoGluon(v0.2.0) [https://auto.gluon.ai/stable/index.html], XGBoost(v0.82), scikit-learn(v0.20.3), Keras(v1.0.8), networkx(v2.0), online MIB server (http://bioinfo.cmu.edu.tw/MIB/structure/), online CD-HIT server (http://weizhong-lab.ucsd.edu/cdhit_suite/cgi-bin/index.cgi?cmd=cd-hit), BLASTP (version 2.7.1+), ImageJ(Java 1.6.0_24), GraphPad(v7.0), Rosetta(2020.08.61146), GROMACS 2019.3 package, PyMOL(v2.4.0a0) The script named contact_prediction.ipynb (from https://github.com/facebookresearch/esm) was used to predict contact probability. Other code used for data analysis is available at https://github.com/wangchulab/MetalNet.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The original protein structure dataset containing 9846 protein sequences and their MSA can be downloaded from https://doi.org/10.1073/pnas.1702664114. Coevolved pairs used in model training can be found in Supplementary Dataset. The proteins and related MSA of prokaryotic species can be downloaded from https://gremlin2.bakerlab.org/db/{species}/fasta/. The Metagenome-pfam MSA and structural model can be downloaded from https://gremlin2.bakerlab.org/db/UNI/. The human spliceosome dataset can be found in Supplementary Dataset. PDB code 6ID0, 6ID1, 6ICZ and 6QW6 were used to construct human spliceosome dataset. We downloaded the information table of protein entities from the PDB server (https://ftp.wwpdb.org/pub/pdb/derived_data/index/entries.idx) to construct unbiased dataset when comparing methods.
UniProt (https://ebi10.uniprot.org) profiles (Date: 20210816), Gene Ontology database (https://www.ebi.ac.uk/QuickGO/) and Pfam database (http://pfam.xfam.org/) were used in analysis.The pdbaa database (16 January 2018 release, ftp://ftp.ncbi.nlm.nih.gov/blast/db/pdbaa.tar.gz) are used in BLASTP.
The structures of citX reported in this paper have been deposited in PDB with the accession numbers of the PDB codes 7DCM (determined by single-wavelength anomalous diffraction) and 7DCN (determined by molecular replacement).

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes for experiments were determined to be adequate based on the prior experience and similar experiments reported in the literature.
Data exclusions	We excluded proteins which can not provide enough coevolved pairs for model training. No data were excluded in biological analysis, and the data points are provided in box plots when applicable.
Replication	All experiments were performed in three independent biological replicates unless stated otherwise. All attempts at replication were successful.
Randomization	Allocation was random.
Blinding	We were blinded to group allocation.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging